

Assignment 2.4

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Task 1

Book class for a simple library management module

```
class Book:
```

```
    def __init__(self, title, author):
```

```
        # Initialize book attributes
```

```
        self.title = title
```

```
        self.author = author
```

```
    def summary(self):
```

```
        # Return a short description of the book
```

```
        return f'"{self.title}" by {self.author}'
```

```
# Example usage
```

```
book1 = Book("The Alchemist", "Paulo Coelho")
```

```
print(book1.summary())
```

Output

'The Alchemist' by Paulo Coelho

Task 2

Sorting user records by age (Cursor-style solution)

```
users = [  
    {"name": "Ravi", "age": 25},  
    {"name": "Anita", "age": 19},  
    {"name": "Kiran", "age": 30}  
]  
  
users.sort(key=lambda x: x["age"])  
  
for user in users:  
    print(user)
```

Output

```
{'name': 'Anita', 'age': 19}  
{'name': 'Ravi', 'age': 25}  
{'name': 'Kiran', 'age': 30}
```

Task 3

Simple Calculator Using Functions

```
def add(a, b):  
    return a + b
```

```
def subtract(a, b):  
    return a - b
```

```
def multiply(a, b):  
    return a * b
```

```
def divide(a, b):  
    if b == 0:  
        return "Division by zero not allowed"  
    return a / b
```

Main program

```
print("Simple Calculator")  
print("1. Add")  
print("2. Subtract")  
print("3. Multiply")  
print("4. Divide")
```

```
choice = int(input("Enter choice (1-4): "))
```

```
num1 = float(input("Enter first number: "))
num2 = float(input("Enter second number: "))

if choice == 1:
    print("Result:", add(num1, num2))
elif choice == 2:
    print("Result:", subtract(num1, num2))
elif choice == 3:
    print("Result:", multiply(num1, num2))
elif choice == 4:
    print("Result:", divide(num1, num2))
else:
    print("Invalid choice")
```

Output

```
Enter choice (1-4): 1
Enter first number: 10
Enter second number: 5
```

Result: 15.0

Task 4

Armstrong Number Check (Basic Version)

```
num = int(input("Enter a number: "))
```

```
sum_val = 0
```

```
temp = num
```

```
digits = len(str(num))
```

```
while temp > 0:
```

```
    digit = temp % 10
```

```
    sum_val += digit ** digits
```

```
    temp //= 10
```

```
if sum_val == num:
```

```
    print("Armstrong number")
```

```
else:
```

```
    print("Not an Armstrong number")
```

Output

Enter a number: 153

Armstrong number